COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-016: Network Routing and SD-WAN Routers

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Title	Network Routing and SD-WAN Routers
Executive Sponsor	Network Services Managers
Issuing Authority	Office of Information Technology (OIT)
Applies to	OIT, state agencies as defined in C.R.S. 24-37.5-101(28), and vendors contracted by OIT or its stakeholders.
Tier	Standard
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1. Purpose

This standard is issued under the authority of the Governor's Office of Information Technology (OIT) (C.R.S. 24-37.5, et seq.) and the Office of Enterprise Architecture (Rule 8 CCR 1501-7). It communicates the standards OIT has developed for enterprise-standard configurations of network routers, describing the minimum requirements for network routers.

2. Scope

These standards apply to OIT, state agencies as defined in C.R.S. 24-37.5-101(28), and vendors contracted by OIT or its stakeholders, to enforce technology standardization and assist in making the use, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews and approves the Standard (this document).

Network Services Managers

- Owner of Standard; engages stakeholders and subject matter experts to establish the standard.
- Updates document standards and version number; presents to the Architecture Review Board for approval and adoption.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Due to the varying scale of agency networks, building complexities, and diverse connectivity requirements, a spectrum of common routing architectures is outlined. Routers play a pivotal role in directing traffic between different networks, ensuring efficient data flow, and providing secure connectivity for integration with the Colorado State Network (CSN) and SD-WAN.

Mandatory Core and Edge Router Features

- Support for IPv4 and IPv6 routing, static routes, and advanced IP routing protocols (OSPF, BGP, IS-IS).
- Policy-based routing and multicast capabilities, including PIM and IGMP snooping (v1, v2, v3).
- High-capacity interface options for Gigabit, 10 Gigabit, 40 Gigabit, and 100 Gigabit Ethernet ports.
- Redundant power supplies and fan options for high availability.
- Hardware-based forwarding for high-performance packet processing.
- Support for Virtual Routing and Forwarding (VRF-Lite) for network segmentation.
- NetFlow/IPFIX support for traffic analysis and monitoring.

Mandatory SD-WAN Features & Requirements

- FIPS 140-2/3 Validated Encryption: All traffic, especially CJIS and FTI, must be encrypted in transit using FIPS 140-2/3 validated IPsec VPN tunnels (e.g., AES-256).
- Centralized Security Policy Enforcement: Leverage vManage for consistent, enterprise-wide policy definition and deployment, including granular Role-Based Access Control (RBAC).
- Network Segmentation: Support for VRFs and VLANs to logically isolate sensitive data and enforce least privilege principles.
- Integrated Security Services using SASE: Must include an application-aware cloud firewall, IPS, URL filtering, and Advanced Malware Protection (AMP).
- Strong Authentication & Secure Management: Support for TACACS+, RADIUS, SSHv2, and SNMPv3 for administrative access.
- Comprehensive Logging & Auditing: Generate detailed, tamper-resistant audit logs integrated with centralized SIEM systems.
- High Availability & Resilience: Support dual-path redundancy and dynamic path selection to ensure continuous connectivity.

Mandatory Security Features for All Router Types

- Management connectivity via SSHv2 and SNMPv3; individual administration logins; support for TACACS+ and RADIUS servers.
- Capability to store Type 9 hashed passwords using the SCRYPT hashing algorithm (RFC 7914); RSA key generation of at least 2048 bits.
- Support for session/login timeouts; Access Control Lists (ACLs) for traffic filtering; Control Plane Policing (CoPP); and Unicast Reverse Path Forwarding (uRPF).
- Event logging to a centralized logging server and NTP support for timestamp accuracy.

Refresh (End of Life)

- Remote Location Edge Routers: 4–6 years recommended refresh cycle.
- Medium Site Edge Routers: 5–7 years recommended refresh cycle.
- Core/Chassis Routers: 7–8 years recommended refresh cycle.
- At a minimum, manufacturers' lifecycle recommendations must be followed, replacing equipment before security vulnerabilities or loss of OS support.

5. Exceptions and Variance

Requests for variances from any provision of this standard must be made to the Director of Network Services for review and approval. The Exception Request must address business or technical reasons for the inability to adhere to one or more requirements, including proposed compensating controls. The Secure Configuration Exception Request form must be submitted to oit_ciso@state.co.us.

6. References

- OIT Hardware Standard Guidance
- NIST SP 800-171, Rev. 3, Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations
- IRS Safeguard Computer Security Evaluation Matrix (SCSEM)
- FIPS 140-2/3, Security Requirements for Cryptographic Modules
- Colorado OIT CISP-004 (Data Security Policy) and CISP-003 (IT Security Risk Management Policy)

7. Definitions

Edge Router: A router located at a network boundary enabling an internal network to connect to external networks.

Core Router: A router optimized for handling high-volume, high-speed data traffic within the backbone of an extensive network.

SD-WAN (Software-Defined Wide Area Network): A virtual WAN architecture allowing enterprises to leverage any combination of transport services to connect users to applications securely.

SASE (Secure Access Service Edge): A cloud-based architecture combining WAN capabilities with comprehensive network security functions into a single, integrated cloud service model.

BGP (Border Gateway Protocol): A standardized exterior gateway protocol for exchanging routing and reachability information among autonomous systems.

Control Plane Policing (CoPP): A feature configuring a QoS policy for the control plane to protect the CPU from excessive traffic.

OSPF (Open Shortest Path First): A widely used interior gateway protocol routing packets within a single autonomous system based on shortest path first.

uRPF (Unicast Reverse Path Forwarding): A security feature that helps mitigate IP spoofing by verifying the source IP address of incoming packets against the router's routing table.